

OVERVIEW OF WELDING PROGRAMS

Applied Research: How the industry has benefited from involvement with the CCWJ

General Industrial Landscape:

- 3018 participants networked among themselves and with global leaders of industry through 61 seminars led by the CCWJ (satisfaction rate was very high at 92%, selected quotes listed in Appendix A)
- 60 businesses and 200 participants learned about latest technologies through the CCWJ-led AMFI program
- Several participating companies disclosed investing in training and in new equipment, and were able to increase their gross margins following participation in a training session at the CCWJ
- CCWJ led trade missions for the 2009, 2013, and 2017 Schweissen & Schneiden trade show in Germany, and hosted an AB function at Fabtech 2011, 2015, 2016. Highly successful trade missions and AB function took place at Fabtech 2015 and 2016, and CanWeld Expo 2016, and similar missions are currently being prepared for 2017 Fabtech.
- CCWJ hosted 3 Industry roundtables including Manufacturing for Nuclear Applications (2021, 16 participants), Alberta Industrial Manufacturing (2018, with Government of Alberta, 39 participants), Advanced Manufacturing (2018, with Western Economic Diversification, 15 participants)
- Memorandum of Understanding between Germany and the Province of Alberta on an Exchange on Welding Technologies
- Access to unique equipment at CCWJ for industry

AltaSteel (<https://www.altasteel.com/>)

- The CCWJ is developing a quantitative model of hydrogen through the complete production cycle of heat-treated, high-carbon grinding and SAG product

Amsted Rail (<https://www.amstedrail.com/>)

- The CCWJ is working with Amsted on developing an innovative, cost and quality advantageous approach to manufacturing and repair railroad wheels

Apollo Clad, Edmonton (<http://apollomachine.com/Apollo-Clad.html>):

- Apollo developed a new outlet for its product based on the MSc project of a CCWJ student (qualifying product through full-scale fatigue testing)
- Apollo funded a PhD project that makes it possible to predict laser cladding parameters ahead of time, which directly translates into time and cost savings. The PhD student involved was hired by Apollo directly after graduation.

Bab Hitachi, Saskatoon (<http://www.pasca.mhps.com/>):

- CCWJ developed a hot-wire process for cladding using proprietary Bab Hitachi equipment
- Bab Hitachi hired a CCWJ graduate to work on their cladding business

CASTI, Edmonton (<https://www.casti.ca>)

- CASTI hired several CCWJ students to create a systematic catalog of welding defects to implement in their training programs

Enbridge Pipeline, Edmonton (<https://www.enbridge.com/>):

- Enbridge is collaborating with the CCWJ to determine the effect of misalignment in pipeline welds on hydrogen cracking to reduce cracking incidents (which account for 15.8% of all pipeline incidents)



Group Six Technologies, Edmonton (<http://groupsixtech.com/>):

- Group Six is collaborating with the CCWJ and other industrial and research partners on an international research project to commercialize a new welding consumable which increases efficiency of laser cladding and decreases costs
- The project opens new markets to the Canadian manufacturing industry and further links Canada in the international R&D network
- The project is a direct outcome of the networking efforts of the CCWJ between Canadian industry, international research institutions, and consumable manufacturers

Syncrude, Alberta (<http://www.syncrude.ca/>):

- Development of tubular wires for overlays. Discovery that carbides were not dissolved but bounced off the surface. Follow-up work on wire development with Stooddy, Arctec, HC Starck
- Discovery that Cr was detrimental in Ni/WC overlays. Industry standard now is matrix material without Cr

Stooddy (<http://victortechnologies.com/stooddy/>):

- Assistance in the development of consumables using high speed video
- Testing of custom made waveforms on power supplies tailored for specific consumables

Wilkinson, Alberta (<http://wilkinsonsteel.ca/>) – now owned by Samuel, Son & Co.:

- Discovery of the role of dilution in carbide fraction in overlays. Development of predictive tools for control.
- Frequent assistance in qualification of product
- Technical assistance in the form of seminars and mentoring

Fundamental Research:

- **Leading the field of technology with latest discoveries from science and equipment:** membership in key committees such as IIW: Assembly National Delegate for Canada; Expert for Commission XII and SG 212; Member of IIW TMB; and AWS: A5G Subcommittee on Hard Facing Filler Metals, EC Education Committee, Higher Education – Engineering Subcommittee, Welding R&D Committee, Standardization in Welding Modelling Committee; MaJIC Board of Directors
- **Award-winning research:** more than 150 awards since 2009, including the AWS Image of Welding: Educational Facility Award (2013) and CWA and AWS fellowships.
- **Provide access to wide range of welding processes and specialized/unique analysis equipment to the industry:** GMAW; GTAW; PTAW; SAW; SMAW; FSW; LBW; laser cladding; Hot-Wire; plasma cutting; high speed video; thermography; complete data acquisition; full metallography lab; hardness mapper (Rockwell, Vickers); gas chromatographer (O,N,H); Charpy / instrumented impact tester; surface profilometer; dilatometer; all levels optical and electron microscopy; ThermoCalc and COMSOL; JMatPro (; Simufact.Welding; TransWeld; VrWeld

High Schools and Young Talent

- We hosted 30 High School internships, and tours for 6 teachers and 454 High School students from St Joseph's High School, J. Percy Page High School, Old Scona Academic High School, Grandview Heights School, Harry Ainlay School, Archbishop MacDonald High School, Ottewell Junior High School, and the WISEST program (since 2016), USchool program (since 2016), and TeamUP (2017).
- The students are exposed to arc welding processes and robotics
- Interns benefit from contacts to industry to get internships in the industry
- Schools benefit from know-how and technologies that are only available at CCWJ



- Free access to our high-speed videos for teaching
- Award-winning interdisciplinary projects attract young talent and expose them to welding: CCWJ outreach program has been successful in getting people, and has won a first prize in the AWS poster competition at Fabtech 2015

Open Labs:

We hosted tours and demonstrations for the general public with 3658 participants at events such as CWA/AWS Open Labs, UofA Open House, and Engineering Expo.

Undergraduate Students

- Total of 232 4-month research projects, 168 individual students since 2010
- 33 summer students and 56
- international visiting students since 2010
- In-depth exposure to welding with a wide range of projects from practical to theoretical through formal classroom teaching and hands-on experience in research
- Possibility to become co-authors in research publications and CWA Journal articles
- Exposure to international welding research
 - Solid understanding of current welding technology
 - Participation in exchange programs and international internships
 - Attend guest lectures by high-profile international visitors
- Exposure to industry
 - Develop understanding of industry needs
 - Help find internships / co-op placements / jobs

Graduate Students

- Multiple streams of studies: MSc in Welding Engineering, MEng in Welding Engineering, PhD, Post-doc
- Two tracks:
 - Leaders in the practical application of advanced welding processes, metallurgy, and automation
 - Leaders in the creation of new technologies in advanced welding processes, metallurgy, and automation
- Gain in-depth understanding of welding processes, science, and needs of industry
- Exposure to international welding research
 - Solid understanding of current welding technology
 - Hands-on welding experience
 - Knowledge: understanding of welding fundamentals
 - Resourcefulness: close familiarity with welding industry and practical considerations
 - Leadership: CWA student chapter executive, CWA Edmonton chapter
 - Flexibility: Able to discriminate important from secondary, myths from truth
 - Communication: Open and clear discussion of ideas with the whole spectrum of welding professionals
 - Team players: collaboration with international researchers and teamwork in the group
 - Exposure: attend trade shows (Fabtech, GPS, etc.) and conferences (e.g. CanWeld, IIW)



Recent Courses offered by Faculty:

Term	Course		Instructor	Size
F14	MATE 466/630	Special topics in Material Engineering: Fundamentals of Welding	Leijun Li / Patricio Mendez	30 UG 2 G
W15	CHE 314	Heat Transfer	Patricio Mendez	110 UG
W15	MATE 202	Mat. Sci. II	Leijun Li	131 UG
W15	MATE 466/694	Advanced Manufacturing	Patricio Mendez / Leijun Li	13 UG 8 GS
W15	CME483	Colloquium	Patricio Mendez	22 UG
W15	CME483	Colloquium	Leijun Li	25 UG
F15	MATE 202	Materials Science	Leijun Li	130 UG
W16	MATE 466/673	Welding Metallurgy	Leijun Li	40 UG
F16	MATE 466/630	Special topics in Material Engineering: Fundamentals of Welding	Patricio Mendez	4 UG 4 G
W17	CHE 314	Heat Transfer	Patricio Mendez	118 UG
F17	MATE 466/630	Special topics in Material Engineering: Fundamentals of Welding	Patricio Mendez	21 UG 5 G
W18	CHE 314	Heat Transfer	Patricio Mendez	63
F18	MATE 466/630	Special topics in Material Engineering: Fundamentals of Welding	Patricio Mendez	18 UG 4 G
W19	CHE 314	Heat Transfer	Patricio Mendez	66 UG
F19	MATE 681	Advanced Welding Engineering	Patricio Mendez	7 UG 13 G
W20	CHE 314	Heat Transfer	Patricio Mendez	58 UG
F20	MATE 681	Advanced Welding Engineering	Patricio Mendez	18 UG 4 G
W21	CHE 314	Heat Transfer	Patricio Mendez	35 UG
F21	MATE 681	Advanced Welding Engineering	Patricio Mendez	8 UG 15 G
W22	CHE 314	Heat Transfer	Patricio Mendez	



CCWJ has a strong focus on Equity, Diversity, and Inclusivity

The CCWJ is a culturally diverse institution, with up to 13 different nationalities and numerous languages present. We actively and successfully focus on providing an equitable, diverse, and inclusive environment, and support our students in thriving as a lab member, and finding a career in engineering, science, and the welding industry. This support takes place on several levels: on a secondary school level we maintain an outreach program to local high-schools on both the applied (St. Joseph's High School) and the theoretical side (Old Scona), which exposes students to welding and engineering and encourages them to consider a career in this area. We also participate in the WISEST program and we have hosted four female interns and three female summer students. When the economic downturn in Alberta made it difficult to find employment in welding for one of the female welding apprentices who worked with our lab, we used our contacts to industry and research to ensure she finds welding-related employment that allows her to build her resume,



expand her experiences, and join the welding community as a successful professional.

We recognize that we are still facing challenges to improve our numbers of female students and indigenous students, and we are making active efforts and seek ongoing feedback on the effectiveness of these efforts. Faculty, staff, and students regularly attend workshops to ensure that the lab environment is inclusive and welcoming to everybody (including an indigenous awareness session in 2021), and regularly engage external resources to adjust our strategies.



Description of graduate placement

Apollo Clad (<https://apollomachine.com/Apollo-Clad.html>)

- The head of R&D (Doug Hamre) was hired as a result of CCWJ matchmaking
- Hired Gentry Wood (CCWJ PhD '17) as welding engineer in 2017
- Apollo developed computer models of its cladding and heat treating through an intern hired from the CCWJ (Satya Gajapathi, MSc '11)
- Hired Ata Kamyabi (PhD '15) as Welding R&D Engineer in 2019
- Hired Dakota Jones (MEng '18) as R&D Engineer in 2018

AITF (<https://www.albertatechfutures.ca/>):

- Hired Thilan Liyanage (CCWJ MSc '10) as a lead engineer in their welding and surfacing division

Areva (<http://www.areva.com/>):

- Hired Julien Chapuis (CCWJ PostDoc '12) as the head of their Head of Welding Processes Unit, Areva NP, France



BMT (<https://www.bmt.org/>)

- Hired Stuart Guest (CCWJ PhD '14) as Sr. Materials & Welding Specialist in 2021

CCNB (<https://www.ccnb.ca>)

- Hired Satya Gajapathi (CCWJ MSc '11) as Innovation Lead, Advanced Manufacturing and Welding at CCNB-INNOV in 2017

Emerson Automation Solutions

- Hired Kevin Scott (CCWJ MSc '11) as Welding Engineer in 2014

eni saipem (<http://www.saipem.com>):

- Hired Shahzad Nadeem (CCWJ MEng '12) as an engineer in their Materials, Corrosion and Welding Engineering Department in Canada
- Hired Ivan Le Gall (CCWJ visiting student '14) as Welding Engineer

Group Six (<https://groupsixtech.com/>):

- Chief R&D engineer Steven Borle (CCWJ MSc '14) was hired straight after his MSc at the CCWJ to create and run Group Six metallurgical laboratory
- Group Six hired Matthew Dewar (CCWJ MSc '12) to lead their robot and automation programming
- Hired Syed Alam (CCWJ BSc '18) for 2 work terms
- Hired Avirup Bose (CCWJ Intern '20) for 2 work terms

IRIS NDT (<https://www.irisndt.com/ca/>):

- Matthew Bell furthered his education with an MSc at the CCWJ, which he completed in 2018
- De-ann Perrey (nee Rollings, CCWJ MEng '14) joined IRIS NDT in Calgary as a Consulting Engineer
- Hired Kurtis Bell (CCWJ MSc '18) as Welding Consulting Engineer in 2017

Indalco Alloys / Lincoln (<http://www.indalco.com/>):

- Set up a special project to train and hire Cory McIntosh (CCWJ MSc '11) as Materials Engineer

ITW / Miller (<https://www.millerwelds.com/>):

- Hired Kevin Scott (CCWJ MSc '11) as Arc Research Scientist for their Global Welding Technology Center in Appleton, WI straight after his MSc at the CCWJ in 2011

Lincoln Electric Canada

- Hired Vivek Sengupta (CCWJ MSc '17) as Welding Engineer in 2018, and promoted him Consumable R&D and Weld Tech Manager for Lincoln Canada in 2021

Nuclear Waste Management Organization (<http://www.nwmo.ca>):

- Hired Thilan Liyanage (CCWJ MSc '10) as Materials, Weld, Coating, and Inspection Specialist

Qualimet

- Hired Cory McIntosh (CCWJ MSc '17) as Welding Engineer in 2021

Stantec (<https://www.stantec.com/>):

- Hired Stuart Guest (CCWJ PhD '14) as a lead engineer in their welding consulting business, which is expanding and has requested more interns from the CCWJ.
- Hired Cory McIntosh (CCWJ MSc '17) as Junior Integrity Specialist in 2018

Supreme Steel

- Hired Nairn Barnes (CCWJ PhD '17) as Welding Engineer



Refereed Publications

1. Oles Sendetskyi, M. S., Patricio Mendez, Michael Fleischauer, *Conflat Cell for in-Situ Electrochemical X-Ray Studies of Lithium-Ion Battery Materials in Commercially-Relevant Conditions*. Journal of Applied Crystallography 2021. Under review.
2. **Grams, M. R.** and Mendez, P. F., *Scaling Analysis of the Thermal Stress Field Produced by a Moving Point Heat Source in a Thin Plate*. Journal of Applied Mechanics, 2021. 88: p. 011001, 1-10.
3. **Grams, M. R.** and Mendez, P. F., *A General Expression for the Welding Tendon Force*. Journal of Manufacturing Science and Engineering, 2021. 143 (12): p. 121002-1, 121002-12.
4. **Hintze Cesaro, A.** and Mendez, P. F., *Effect of the Heating Rate on Austenite Formation in Low Carbon Microalloyed Steels*. Welding Journal, 2021. In print.
5. **Lu, Y.** and Mendez, P. F., *Characteristic Values of the Temperature Field Induced by a Moving Line Heat Source*. International Journal of Heat and Mass Transfer, 2021. 166: p. 120671, 1-15.
6. Molina, C., Araujo, A., **Bell, K.**, Mendez, P. F., and Chapetti, M., *Fatigue Life of Laser Additive Manufacturing Repaired Steel Component*. Engineering Fracture Mechanics, 2021. 241: p. 107417, 1-11.
7. **Velázquez-Sánchez, A.**, Delgado-Álvarez, A., Mendez, P. F., Murphy, A. B., and Ramírez-Argáez, M. A., *Dominant Heat Transfer Mechanisms in the GTAW Plasma Arc Column*. Plasma Chemistry and Plasma Processing, 2021. 41: p. 1497-1515.
8. **Lu, Y., Wang, Y.**, and Mendez, P. F., *Width of Thermal Features Induced by a 2-D Moving Heat Source*. International Journal of Heat and Mass Transfer, 2020. 156: p. 119793, 1-14.
9. **Delgado-Álvarez, A.**, Mendez, P. F., Murphy, A. B., and Ramírez-Argáez, M. A., *Generalized Representation of Arc Shape, Arc Column Characteristics and Arc-Weld Pool Interactions for DC Electric Arcs Burning in Monoatomic Gases*. Journal of Physics D: Applied Physics, 2020. 54 (5): p. 055001, 1-12.
10. **Delgado-Álvarez, A.**, Mendez, P. F., and Ramírez-Argáez, M. A., *Dimensionless Representation of the Column Characteristics and Weld Pool Interactions for a DC Argon Arc*. Science and Technology of Welding and Joining, 2019.
11. **Sengupta, V., Havrylov, D.**, and Mendez, P., *Physical Phenomena in the Weld Zone of Submerged Arc Welding—a Review*. Welding Journal, 2019. 98 (10): p. 283S-313S.
12. **Wang, Y., Lu, Y.**, and Mendez, P. F., *Scaling Expressions of Characteristic Values for a Moving Point Heat Source in Steady State on a Semi-Infinite Solid*. International Journal of Heat and Mass Transfer, 2019. 135: p. 1118-1129.
13. **He, H.**, Gou, W., Lin, S., Yang, C., and Mendez, P., *GTA Weld Brazing a Joint of Aluminum to Stainless Steel*. Welding Journal, 2019. 98 (12): p. 365s-378s.
14. **He, H.**, Gou, W., Wang, S., Hou, Y., Ma, C., and Mendez, P. F., *Kinetics of Intermetallic Compound Layers During Initial Period of Reaction between Mild Steel and Molten Aluminum*. International Journal of Materials Research, 2019. 110 (3): p. 194-201.
15. Mendez, P. F., **Wang, Y.**, and **Lu, Y.**, *Scaling Analysis of a Moving Point Heat Source in Steady-State on a Semi-Infinite Solid*. Journal of Heat Transfer, 2018. 140: p. 081301-9.
16. **Barnes, N.**, Clark, S., Seetharaman, S., and Mendez, P. F., *Growth Mechanism of Primary Needles During the Solidification of Chromium Carbide Overlays*. Acta Materialia, 2018. 151: p. 356-365.
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18. **McIntosh, C.** and Mendez, P. F., *Experimental Measurements of Fall Voltages in Gas Metal Arc Welding*. Welding Journal, 2017. 96: p. 121s-132s.
19. **Sengupta, V.** and Mendez, P. F., *Effect of Current on Metal Transfer in SAW. Part 2: AC*. Welding Journal, 2017.



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20. **Sengupta, V.** and Mendez, P. F., *Effect of Current on Metal Transfer in SAW. Part 1: DCEP.* Welding Journal, 2017. 96: p. 241s-249s.
21. **Sengupta, V.** and Mendez, P. F., *Effect of Fluxes on Metal Transfer and Arc Length in SAW.* Welding Journal, 2017. 96: p. 334s-353s.
22. **Barnes, N., Borle, S.,** and Mendez, P. F., *Large Anomalous Features in the Microstructure of Chromium Carbide Weld Overlays.* Science And Technology Of Welding And Joining, 2017. 22 (7): p. 595-600.
23. **Kamyabi Gol, A., Herath, D.,** and Mendez, P. F., *A Comparison of Common and New Methods to Determine Martensite Start Temperature Using a Dilatometer.* Canadian Metallurgical Quarterly, 2016. 56 (1): p. 85-93.
24. **McIntosh, C., Chapuis, J.,** and Mendez, P. F., *Effect of Ar-Co₂ Gas Blends on Droplet Temperature in GMAW.* Welding Journal, 2016. 95 (8): p. 273s-279s.
25. **Kamyabi-Gol, A., Clark, S. J., Gibbs, J. W.,** Seetharaman, S., and Mendez, P. F., *Quantification of Multiple Simultaneous Phase Transformations Using Dilation Curve Analysis (DCA).* Acta Materialia, 2016. 2016: p. 231-240.
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27. **Wood, G.** and Mendez, P. F., *Disaggregated Metal and Carbide Catchment Efficiencies in Laser Cladding of Nickel-Tungsten Carbide.* Welding Journal, 2015. 94 (11): p. 343s-350s.
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30. Mendez, P. F., Goett, G., and **Guest, S. D.,** *High-Speed Video of Metal Transfer in Submerged Arc Welding.* Welding Journal, 2015. 94 (10): p. 325s-332s.
31. **Gibbs, J. W.,** Schlachter, C., Mayr, P., **Kamyabi-Gol, A.,** and Mendez, P. F., *Cooling Curve Analysis as an Alternative to Dilatometry in Continuous Cooling Transformations.* Materials Transactions A, 2015. 46A (1): p. 148-155.
32. **Wood, G., Islam, S. A.,** and Mendez, P. F., *Calibrated Expressions for Welding and Their Application to Isotherm Width in a Thick Plate.* Soldagem & Inspecao, 2014. 19 (3): p. 212-220.
33. Gerlich, A. P., **Izadi, H.,** Bundy, J., and Mendez, P. F., *Characterization of High Strength Weld Metal Containing Mg-Bearing Inclusions.* Welding Journal, 2014. 93 (1): p. 15s-22s.
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39. **Gajapathi, S. S.,** Mitra, S. K., and Mendez, P. F., *Part 1: Development of New Heat Source Model Applicable to Micro Electron Beam Welding.* Science And Technology Of Welding And Joining, 2012. 17 (6): p. 429-434.
40. Mendez, P. F., *Synthesis and Generalisation of Welding Fundamentals to Design New Welding Technologies:*



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41. **Lehnhoff, G.** and Mendez, P. F., *Scaling of Non-Linear Effects in Heat Transfer of a Continuously Fed Melting Wire. International Journal of Heat and Mass Transfer*, 2011. 54: p. 2651-2660.
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43. **Soderstrom, E., Scott, K. M.**, and Mendez, P. F., *Calorimetric Measurement of Droplet Temperature in GMAW. Welding Journal*, 2011. 90 (4): p. 77s-84s.
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46. Mendez, P. F., **Tello, K.**, and Lienert, T. J., *Scaling of Coupled Heat Transfer and Plastic Deformation around the Pin in Friction Stir Welding. Acta Materialia*, 2010. 58 (18): p. 6012-6026.
47. **Tello, K. E.**, Gerlich, A. P., and Mendez, P. F., *Constants for Hot Deformation Constitutive Models for Recent Experimental Data. Science and Technology of Welding and Joining*, 2010. 15 (3): p. 260-266.
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50. **Tordonato, D. S., Madeni, J. C.**, Babu, S. S., Liu, S., and Mendez, P. F., *A New Method for the Design of Welding Consumables. Welding Journal*, 2010. 89 (10): p. 201s-209s.

Recent Awards & Recognitions

Students

2021	Alejandro Hintze	AWS Fellowship (renewal)
		Tenaris Roccha Fellowship Renewal
	Amanda Dubrule	Ila and John Mellow Prize
		Philosophy Senior Student Award in recognition of your academic success in philosophy in the past academic year
	Carter Trautmann	CWB-A Edmonton Graduate Annual Award for Excellence in Welding Engineering
	Mackenzi Johnston	2020-2021 Leaders in Equality Award of Distinction Program, Women in STEM Award
	Mitchell Grams	IIW Future Leader
	Rishiekesh Ramgopal	University of Alberta Graduate Recruitment Scholarship
	Wesley Lesburg	CWB-A Edmonton Undergraduate Annual Award for Excellence in Welding Engineering
2020	Xinrui Liu	China Scholarship Council (CSC) Scholarship
		Mitacs Graduate Studies Scholarship
	Alejandro Hintze	AWS Fellowship (renewal)
		Tenaris Roccha Fellowship Renewal
	Daniel Ramirez	CWB-WF National UofA Graduate Scholarship for Excellence in Welding Engineering
	Mackenzi Johnston	AWS District Scholarship
		AWS Jerry Hope Endowment Scholarship



2019		CWB-A Edmonton Undergraduate Annual Award for Excellence in Welding Engineering
		A. Louise McKinney Post-Secondary Scholarship
		Albert Hanson Memorial Scholarship in Materials Engineering
	Mitchell Grams	IPC Best Paper (Undergraduate / Graduate Student), International Pipeline Conference (IPC) 2020
	Syed Alam	CWB-A Edmonton Graduate Annual Award for Excellence in Welding Engineering
		CWB-WF National UofA Graduate Scholarship for Excellence in Welding Engineering
	Alejandro Hintze	AWS Fellowship
		CWB-WF National UofA Graduate Scholarship for Excellence in Welding Engineering
		Tenaris Rocca Fellowship
	Alysen Townsley	AWS District Scholarship
	Dmytro Havrylov	CWB-WF National UofA Graduate Scholarship for Excellence in Welding Engineering
	Julie Song	CWB-A Edmonton Undergraduate Annual Award for Excellence in Welding Engineering
	Kyle Foster	CWB-A Edmonton Graduate Annual Award for Excellence in Welding Engineering
	Mehera Salah	Loran Scholar
2018	Mitchell Grams	AWS District Scholarship
		Future Leader Program, Board of Directors AWS
	Yi Lu	AWS District Scholarship
		China Institute Student Travel Scholarship
	Ying Wang	AWS District Scholarship
		AWS Leadership Symposium Scholarship
		China Institute Student Travel Scholarship
	Alejandro Hintze	AWS Jerry Hope Endowment Scholarship
	Kevin Wang	CWB-A Edmonton Undergraduate Annual Award for Excellence in Welding Engineering
	Mitchell Grams	CWB-A Edmonton Graduate Annual Award for Excellence in Welding Engineering
2017		Westmoreland Coal Company Graduate Scholarship
	Syed Alam	AWS District Scholarship
	Yi Lu	CWB-WF National UofA Graduate Scholarship for Excellence in Welding Engineering
		Neil McEwen Memorial Graduate Scholarship in Welding (Materials) Engineering
	Ying Wang	CWB-WF National UofA Graduate Scholarship for Excellence in Welding Engineering
		IIW Travel Bursary Bali
		TW Fraser and Shirley Russell Teaching Fellowship
	Zhaoyang Yan	China Scholarship Council (CSC) Scholarship
	Alejandro Hintze	Captain Thomas Farrell Greenhalgh Memorial Graduate Scholarship
	Dmytro Havrylov	Captain Thomas Farrell Greenhalgh Memorial Graduate Scholarship
	Huan He	China Scholarship Council (CSC) Scholarship
	Mitchell Grams	CWB-WF National UofA Graduate Scholarship for Excellence in Welding Engineering
	Rebekah Bannister	CWB-WF National UofA Graduate Scholarship for Excellence in Welding Engineering
	Syed Alam	CWB-A Edmonton Undergraduate Annual Award for Excellence in Welding Engineering
	Yi Lu	CWB-A Edmonton Graduate Annual Award for Excellence in Welding Engineering
		FGSR Profiling Alberta's Graduate Students Award



	Ying Wang	FGSR Profiling Alberta's Graduate Students Award
		DB Robinson Graduate Scholarship

Faculty

2016 AWS Charles H. Jennings Award, for most valuable contribution to the welding literature ("Primary Chromium Carbide Fraction Control with Variable Polarity SAW." Welding Journal. 94(1). pp. 1s-7s)

2015 AWS Warren F. Savage Memorial Award to the "best paper on metallurgical principles related to welding" published in the Welding Journal Research Supplement ("Characterization of High Strength Weld Metal Containing Mg-Bearing Inclusions." Welding Journal, 2014. 93 (1): p. 15s-22s.)

2015 Undergraduate Research Initiative University of Alberta Award for Outstanding Mentorship in Undergraduate Research & Creative Activities for "(having) gone above and beyond to help support the development of undergraduate researchers."

2014 ASM M. Brian Ives Lectureship Award to "an individual who has made distinguished and significant contributions to the Canadian materials community."

2014 AWS Fellow for a "career of significant achievements in the technical and research arenas."

2014 AWS Adams Memorial Membership Award for "outstanding teaching activities in undergraduate and postgraduate engineering."

2014 University of Alberta Student Union Award for Leadership in Undergraduate Teaching for "excellence in teaching by faculty members who make outstanding contributions in their roles as undergraduate instructors."

2013 CWA Fellowship Award for "exemplary reputation and service to advancements of welding sciences, technology application, research, education, publication of papers, books, journal articles and peer recognition."

2013 AWS William Irrgang Memorial Award "to the individual who had done the most to enhance AWS's goal of advancing the Science and Technology of Welding over the last five years."

2013 AWS Image of Welding: Educational Facility Award to "an educational facility that conducts welding education or training that has demonstrated an exemplary commitment to furthering the Image of welding"

2012 IIW Kenneth Easterling Best Paper Award to "the best contribution made over the three years proceeding on the advancement of knowledge or practice in respect of mathematical modelling of weld phenomena." (OMS: A Computer Algorithm to Develop Closed-Form Solutions to Multicoupled, Multiphysics Problems in Proceedings of 10th International Seminar Numerical Analysis of Weldability, p. 219-254. Graz-Seggau, Austria)

2011 AWS William Spraragen Award to the best paper published in the Welding Journal Research Supplement ("A New Method for the Design of Welding Consumables". Welding Journal, 2010. 89 (10): p. 201s-209s).



Equipment

Characterization	Bruker Galileo G8	Welding	ThermoCalc
	Buehler EcoMet Autopolisher		Transvalor Transweld
	Buehler SimpliMet 1000 Hot Mount		vrWeld
	Cavitar Cavilux HF 810nm 500W laser		VXelements
	Creaform MetraScan 750 3D Scanner		Alicat Scientific high-precision mass flow controllers
	DBO Bifurcation Optics		Cyclomatic Arc Pattern Controller 90A
	Fischione 110 Electro-Polisher		Diffusible Hydrogen Clamp
	Flir A6751sc Thermal Camera		Firefly Orbital Welding System
	Hardness Mapper - Vickers		Furnace - Carbolit Chamber Furnace #1
	Hardness Tester - Wilson Rockwell 2000		Hitachi Hotwire
	Instron Instrumented Impact Tester CEAST 9350		HL-WD Dental Spot Welder
	Linseis Dilatometer L78 R.I.T.A.		Indumatic FSW
	NI DAQ System USB-6351 #1		KUKA KR16 / DKP-450
	Nikon Eclipse MA200 Inverted Microscope		Laserline LDF9000-40 fiber-coupled diode laser, 9kW, with wavelength selector, 2-colour pyrometer
	Nikon SMZ 745T Stereomicroscope		Laserline OTZ 2-axis zoom optics with spot sizes from 5x5mm to 32x32mm
	Ocean Optics Spectrometer		Liburdi Hotwire MIG System
	Phantom miro High Speed Camera		Lincoln /Fanuc Robot
	Phantom V 210 High Speed Camera		Lincoln ACDC 1000 + Gantry
	Phantom VEO1310 High Speed Camera		Lincoln S500 with variable polarity module
	Presi Mecatome T180 Micro Cutter (Diamond Saw)		Lincoln Sculptprint WAAM
	Primes BM+ Beam Profiler		Miller Dynasty 700
	Primes FM+ Beam Profiler		Miller Millermatic 212
	Primes Power Meter CPM-F10		Miller MSW Spotwelder
	Roper Whitney Sample Punch		Miller PipePro 450RFC
	Sharp Broach		Miller Spectrum 2050
	Sharp Broach Ultra Hard		Miller SyncroWave 200
	Struers Labotom 3 Wet Saw		Miller XMT450
	Vulcan 3-550 Furnace		Mini Lathe King KG-8712MI
	XIRIS Camera		Plasmastar PTAW with 230 MV and 400 MV torch
	XIRIS Weld Mic		Precitec YC52 co-axial cladding head, with additional off-axis feeder
	Zygo Zegage Plus Surface Profilometer		Stud Welder TRW
Computer / Software	COMSOL		Thermach AT-1210 Powder Feeder
	Geomagic		ThermalArc SP4000W / 500SP
	Hexagon Simufact Forming		Thermocouple Spotwelder
	Hexagon Simufact Welding		Tormach PCNC 770M CNC
	JMatPro		Yaskawa Motoman Robot
	KUKA PRC		
	Octopuz		
	Rhino 6.0		
	RoboDK		



Appendix A: Quotes from our participants / Outreach Activities

AMFI:

"Great educational material and well presented. Excellent knowledge of materials, great presenters, great discussions throughout [the] workshop [...]" *Ian Cotnam, Metal Alloy*

"[...] vast knowledge of presenters with industry experience. [...] Eye opening what is taking place behind the scenes." *Tim MacIntyre, Suncor*

"Large amount of knowledge in short amount of time – very efficient." *Will Morlidge, Exchanger Industries*

"[Most valuable of the lab tour was] seeing progress being made in bring[ing] welding understanding into [the] 21st century in Alberta. Good on you all! Proud of what Alberta is doing." *Mel Meunier, Aero Trailer*

Outreach:

"The clips that you have on YouTube are fantastic and obviously more up to date than the footage that I currently use." *Darren Jefferies, Senior Welding Inspection lecturer, TWI Ltd*

"Thanks again for providing these clips. It will make my job easier." *Mark Stathers, Okanagan Welding Dept. Okanagan College*

"Thanks for the openness to share findings, video, etc.!" *Mark Robinson, Alberta Pipe Trades College*

CWA/CWB:

"Events hosted by CCWJ or AMFI provide opportunity to network with Industry partners with the same passion to invest and improve our welding industry. It is great exposure to showcase updates and initiatives of our company." *Ken McKen, CWB*

Classes:

"Excellent course. Learned tonnes of valuable and relevant information related to welding. Prof was excellent and the course was well organized." *(MAT E630/466, 2012)*

"General course quality is really good and helps me in my work as well, really an awesome class." *(MAT E630/466, 2012)*

"I will definitely recommend this course to everyone I know. It is a good welding background course. Now I can have a decent conversation about welding during the ASM dinners!" *(MAT E630/466, 2012)*

"This is the best course that I have ever taken" *(MAT E630/466, 2013)*

"Prof. Mendez is so dedicated to teach student with the highest quality and integrity. He did everything to help each student understand the topics very well." *(MAT E630/466, 2013)*

"[...] the quality of information provided was very high. This was the most practical and useful university course I have taken." *(MAT E630/466, 2011)*

"I REALLY enjoyed the combination of lecture, demonstrations, and hands-on experience. I liked the small class size and the open discussions." *(MAT E630/466, 2011)*

Seminars:

"Wonderful lecture and discussion!" *Ken Bird, Supreme Steel*

Internships:

"The experience at the CCWJ will be, in my mind, the most enriching of my education." *Simon Pohardy, Polytechnic Nantes*





Visit by GALD delegation, China, 2015



Visit J.Percy Page High School 2015 (Right)



Seminar on Duplex / Superduplex
Stainless Steels 2014



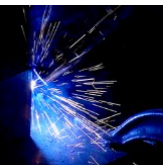
CWA Lab Tour 2014 (Right)



Seminar on Wear Resistant Overlays 2013



AMFI Lab Tour 2014 (Right)



6/12/2015 High school student volunteers in Canadian Centre for Welding and Joining - Department of Chemical & Materials Engineering - University of Alberta
Department of Chemical and Materials Engineering

[Faculty of Engineering](#)

High school student volunteers in Canadian Centre for Welding and Joining

By Nicole Basaraba August 30, 2012

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Edmonton—Not many students entering grade 11 know what they want to study when it comes time to apply to university, but the opposite is true for Sebastian Aravena.

This summer, Aravena, a student from Archbishop MacDonald high school, spent a month volunteering and being trained in the [Canadian Centre for Welding and Joining](#) (CCWJ) in the Department of Chemical and Materials Engineering.

He organized samples of titanium welds by date for Dr. Patricio Mendez and also set up the high-speed cameras for recording experiments. Working with undergraduate student, Nairn Barnes, he learned about the 400A Plasma Transferred Arc Welding (PTAW) system. Aravena observed and set up the camera for Barnes who experimented with the distance between the torch slide and the metal sample to see the effects on the weld and find the optimal setting.

"It was nice to see what they do here, what engineering is all about. It was fun to be with the grad students and to see the university environment. I hope to see more of the university environment this fall," Aravena said.

When he returns to a part-time paid position in the lab this fall, a graduate student will show him how to operate the machines. He will fulfill the safety requirements before he can get his hands on a machine such as the ARC Mate 100IC-series welding robot, which he has been eyeing.

Aravena, not having the textbook background like the university students in the lab, was fresh for molding and Dr. Mendez notes, "The advantage of [Sebastian] not having to be responsible in delivering a research task, is that he is free to join and help everybody. He gets to see the scientific method in action."

<http://www.cme-engineering.ualberta.ca/NewsEvents/CME/News/2012/August/HighSchoolStudentVolunteersinCanadianCentreforWeldingandJoining.aspx> 1/2

6/12/2015 High school student volunteers in Canadian Centre for Welding and Joining - Department of Chemical & Materials Engineering - University of Alberta
Doing well in his math and science classes, Aravena was considering studying medicine or engineering. After his experience in CCWJ, he plans to apply to study Mechanical Engineering, but that wasn't the only thing contributing to his decision.

"Well, I watch MythBusters," he says with a laugh, "and Grant [Imaraha] makes robots and I would really like to make robots for different purposes."

Aravena's advice to his peers who may be trying to choose a post-secondary program is to focus on what they ultimately want to do.

"I think it's good to find a way to experience a career before they pick [their field of study]. The hands-on experience would be useful so that they don't risk not liking it when they're in a program."

Dr. Mendez was pleased with Aravena's work. "Sebastian is so talented. I wanted to explore for the first time having a high school student in the research lab. He accepted the challenge that wasn't perfectly described [to him in advance] and had to use his own judgment to place missing pieces in the correct way."

In addition to the current graduate students and Aravena, the [CCWJ](#) typically has five to ten Dean Research Award undergraduate students and visiting students, this summer it was Masters student, Shataro Yamashita, from the University of Hiroshima, Japan.

The CCWJ holds Open Labs, which present an opportunity for students to get a tour and see demonstrations of the machines in use. The next Open Labs will be on September 14, 2012.

For more information on CCWJ, visit the website at: <http://www.ualberta.ca/~ccwj/>

<http://www.cme-engineering.ualberta.ca/NewsEvents/CME/News/2012/August/HighSchoolStudentVolunteersinCanadianCentreforWeldingandJoining.aspx> 2/2

